

Mathematics A

Unit 2 • Chapter OL-T40 Classified

Cross-session • chapter-organised candidate

11 classified items • 46 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Unit 2 • Chapter OL-T40 • 11 items • 46 marks

Chapter	Items	Marks	Starts
01. OL-T40 Angles in Polygons & Parallel Lines	11	46	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Angles in Polygons & Parallel Lines

Unit 2 • 46 marks • 11 item(s)

June 2025 | 4MA1-1H Q3 | 4 marks

Recover the number of sides from a regular-polygon angle

June 2024 | 4MA1-1H Q5 | 4 marks

Find interior or exterior angle of a regular polygon

June 2023 | 4MA1-2H Q11 | 3 marks

Find interior or exterior angle of a regular polygon

November 2024 | 4MA1-2H Q11 | 5 marks

Combine several regular polygons around lines or a point

June 2022 | 4MA1-1H Q4 | 4 marks

Combine several regular polygons around lines or a point

June 2022 | 4MA1-1H Q23 | 6 marks

Solve an irregular polygon angle or side-count problem

January 2022 | 4MA1-2H Q11 | 4 marks

Find interior or exterior angle of a regular polygon

November 2021 | 4MA1-2H Q5 | 5 marks

Combine several regular polygons around lines or a point

May-Nov 2020 | 1H Q8 | 4 marks

Find the interior-angle sum of a polygon

January 2023 | 1H Q4 | 3 marks

Find the interior-angle sum of a polygon

November 2025 | 1H Q12 | 4 marks

Solve an irregular polygon angle or side-count problem

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

3

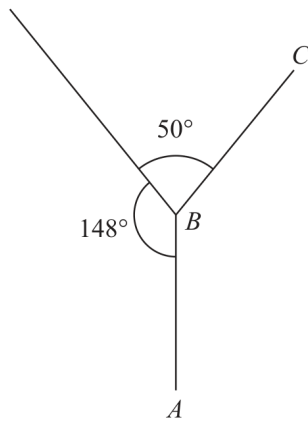


Diagram **NOT**
accurately drawn

AB and BC are two sides of a regular polygon with n sides.

Work out the value of n
Show your working clearly.

$n = \dots\dots\dots$

(Total for Question 3 is 4 marks)

WORKING SPACE

Question 04

4MA1-1H Q5

MODULAR UNIT 2

4 marks

- 5 In the diagram, $ABCDE$ is a regular pentagon.

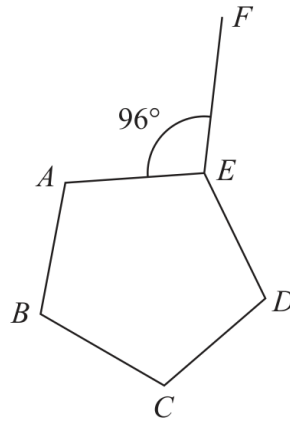


Diagram **NOT**
accurately drawn

Angle $AEF = 96^\circ$

Work out the size of the obtuse angle FED
Show your working clearly.

Working space for Question 04

Continue your method

UNIT 2

4 marks

WORKING SPACE

Question 24

4MA1-2H Q11

MODULAR UNIT 2

3 marks

- 11 Here is a 9-sided regular polygon $ABCDEFGHIJ$, with centre O

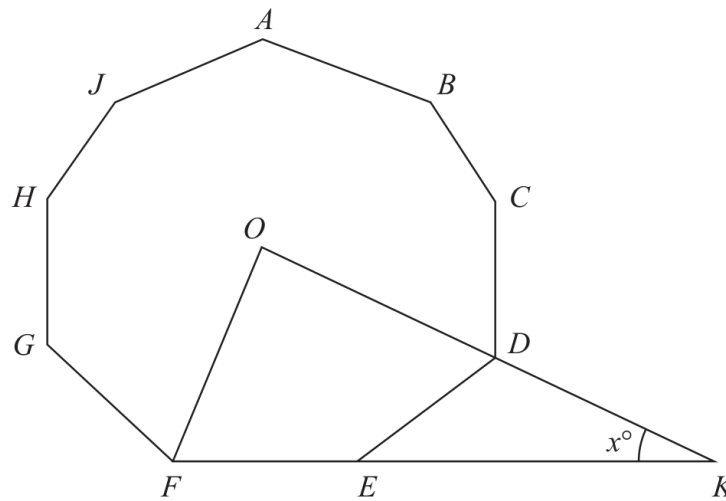


Diagram **NOT**
accurately drawn

ODK and FEK are straight lines.

Work out the value of x

$x = \dots\dots\dots$

Working space for Question 24

Continue your method

UNIT 2**3 marks**

WORKING SPACE

Question 25

4MA1-2H Q11 M01

MODULAR UNIT 2

5 marks

11

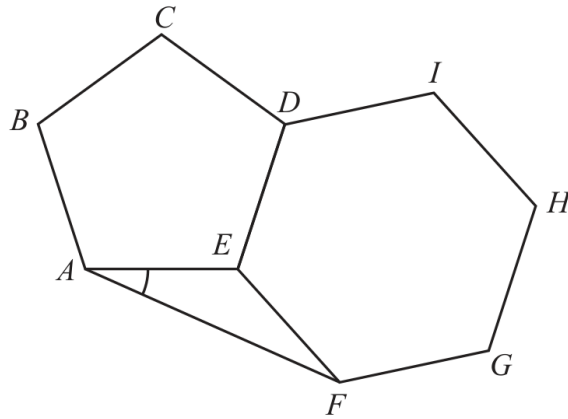


Diagram **NOT**
accurately drawn

$ABCDE$ is a regular pentagon.

$DEFGHI$ is a regular hexagon.

AF is a straight line.

Work out the size of angle EAF

o

Working space for Question 25

Continue your method

UNIT 2

5 marks

WORKING SPACE

Question 05

4MA1-1H Q4 Q4

MODULAR UNIT 2

4 marks

- 4 The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

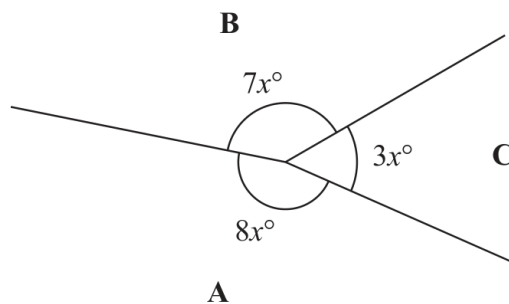


Diagram **NOT**
accurately drawn

Polygon **B** has n sides.

Work out the value of n .

$n = \dots\dots\dots$

WORKING SPACE

Question 19

4MA1-1H Q23 Q23

MODULAR UNIT 2

6 marks

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n

Show clear algebraic working.

Question 23 continued

$n = \dots\dots\dots$

WORKING SPACE

Question 25

4MA1-2H Q11

MODULAR UNIT 2

4 marks

- 11 The diagram shows a regular 10-sided polygon, $ABCDEFGHIJ$

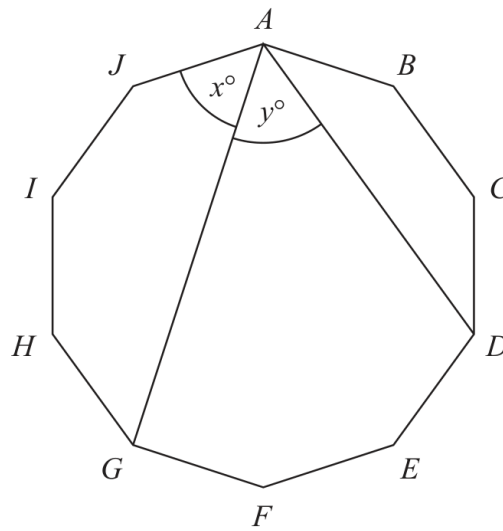


Diagram **NOT**
accurately drawn

Show that $x = y$

Working space for Question 25

Continue your method

UNIT 2

4 marks

WORKING SPACE

- 5 The diagram shows a regular octagon $ABCDHIJK$ and a pentagon $DEFGH$.

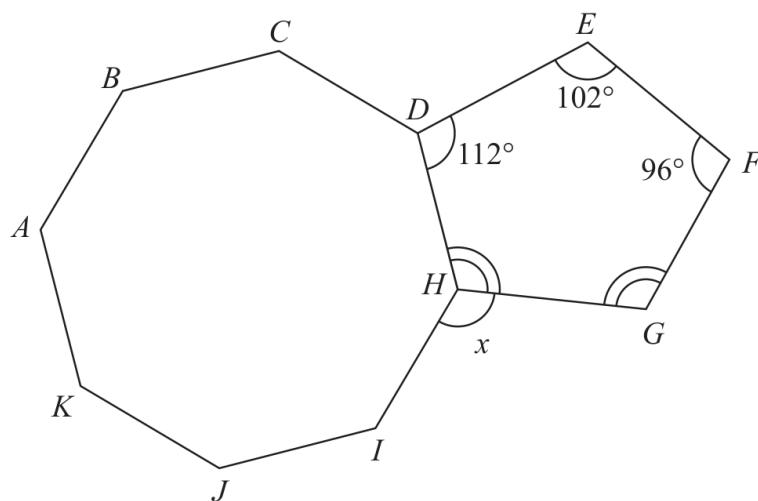


Diagram **NOT**
accurately drawn

Angle GHD = angle FGH .

Work out the size of the angle marked x .
Show your working clearly.

YOUR WORKING

8 Here is a 10-sided polygon.

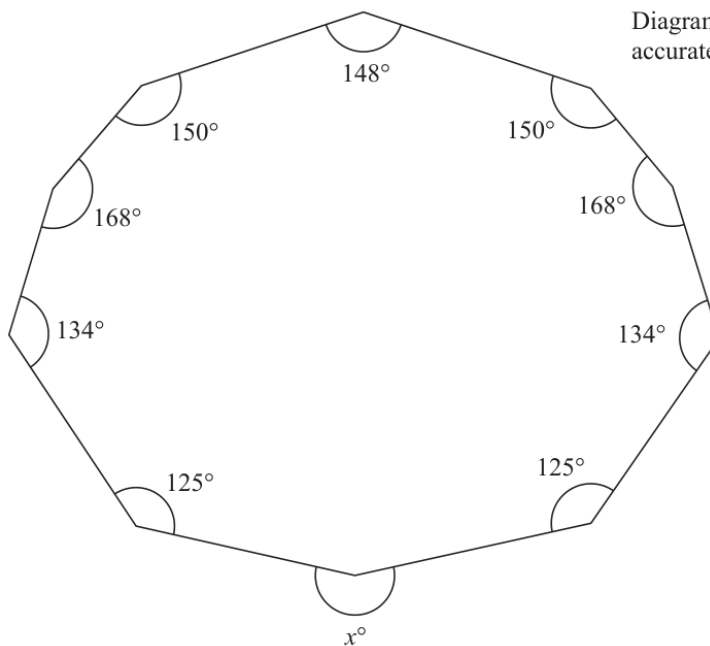


Diagram **NOT**
accurately drawn

Work out the value of x .

$x = \dots\dots\dots$

(Total for Question 8 is 4 marks)

YOUR WORKING

- 4 The diagram shows a pentagon.

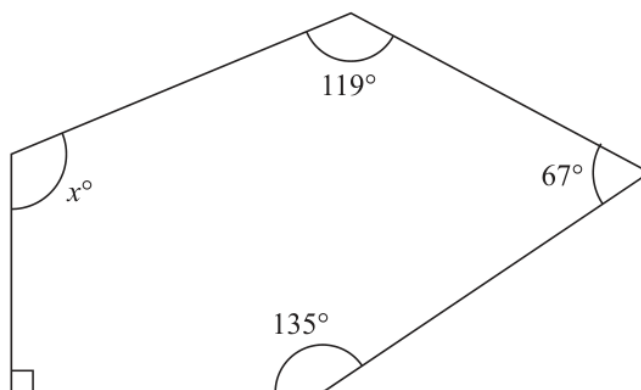


Diagram **NOT**
accurately drawn

Work out the value of x

$x =$

(Total for Question 4 is 3 marks)

YOUR WORKING

12

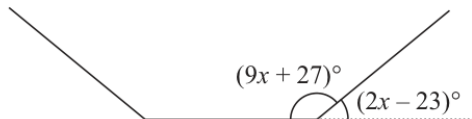


Diagram **NOT**
accurately drawn

The diagram shows part of a regular n -sided polygon with

an interior angle of $(9x + 27)^\circ$

an exterior angle of $(2x - 23)^\circ$

Work out the value of n

$n = \dots\dots\dots$

(Total for Question 12 is 4 marks)

YOUR WORKING
